

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm (New) Examination

University Theory Examination October-2015

MA141.1 Advanced Mathematics

Date: 19/10/2015, Monday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION - I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION - I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION - I is 30 minutes. Answers to SECTION - I must be written during the first 30 minutes of the examination.
3. Answers to SECTION - II and SECTION - III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.
6. Use of scientific non-programable calculator is allowed.

SECTION I

Q.1. Darken the circle of the correct/most suitable choice. [20]

1. The value of $\binom{10}{0}$ is

- (A) 1 (B) 10 (C) 0 (D) 100

2. 7th derivative of function $y = x^4 + x^2 + 1$ is

- (A) $7x^5$ (B) 0 (C) $7x$ (D) 4

3. 3rd derivative of function $y = \cos x$ is

- (A) $\cos x$ (B) $-\cos x$ (C) $-\sin x$ (D) $\sin x$

4. 3rd derivative of function x^3 is

- (A) $9x$ (B) $6x$ (C) 6 (D) 0

5. Let $f(x)$ be a function defined on $[a, b]$ satisfying the following conditions.

- (a) $f(x)$ is continuous on $[a, b]$
(b) $f(x)$ is differentiable on (a, b) .
(c) $f(a) = f(b)$.

Then which of the following is true?

- (A) there exist at least $c \in [a, b]$ such that $f'(c) = a$.
(B) there exist at least $c \in (a, b)$ such that $f'(c) = b$.
(C) there exist at least $c \in (a, b)$ such that $f'(c) = 0$.
(D) there exist at least $c \in [a, b]$ such that $f'(c) = b$.

6. Which of the following condition is not required to be satisfy by function $f(x)$ for Rolle's theorem on $[a, b]$?

- (A) $f(x)$ is continuous $[a, b]$ (C) $f(a) = f(b)$
 (B) $f(x)$ is differentiable on (a, b) (D) $f'(a) = f'(b)$
7. The Laplace transform of $5t$ is
- (A) $\frac{5}{s^2}$ (B) $\frac{5}{s}$ (C) $\frac{1}{5s}$ (D) 5
8. The Laplace transform of e^{-7t} is
- (A) $\frac{1}{s^2 - 49}$ (B) $\frac{1}{s + 7}$ (C) $\frac{1}{s - 7}$ (D) $\frac{1}{s^2 + 49}$
9. The inverse Laplace transform of $\frac{1}{s}$ is
- (A) 1 (B) t (C) t^2 (D) e^t
10. $\mathcal{L}^{-1}[-5f(t)] = \dots$
- (A) $5\mathcal{L}^{-1}[f(t)]$ (C) $\frac{\mathcal{L}^{-1}[f(t)]}{5}$
 (B) $-5\mathcal{L}^{-1}[f(t)]$ (D) $5 + \mathcal{L}^{-1}[f(t)]$
11. Which of the following differential equation is not linear?
- (A) $\frac{dy}{dx} + y = 1$ (C) $\frac{dy}{dx} - xy = xy^2$
 (B) $\frac{dy}{dx} + xy = x^2$ (D) $\frac{dy}{dx} - y \sin x = 1$
12. The order and degree of differential equation $\frac{d^3y}{dx^3} + \frac{dy^2}{dx^2} - y = 0$ are
- (A) order=3 and degree=2 (C) order=2 and degree=1
 (B) order=2 and degree=3 (D) order=3 and degree=1
13. The solution of equation $\frac{dx}{dy} - 5 = 0$ is
- (A) $x = 5y + C$ (C) $x = 5$
 (B) $5x = -y + C$ (D) $y = 5$

14. If $y = 3x - 2$, then differential equation satisfied by y is
- (A) $\frac{d^2y}{dx^2} = 2$ (C) $\frac{d^2y}{dx^2} = -2$
 (B) $\frac{d^2y}{dx^2} = 0$ (D) $\frac{d^2y}{dx^2} = 3$
15. If 1 and 2 are roots of auxiliary equation of second order homogenous differential equation, then solution of that equation is
- (A) $y = C_1e^{-x} + C_2e^{2x}$ (C) $y = C_1e^x + C_2e^{2x}$
 (B) $y = C_1e^{2x} + C_2e^{-x}$ (D) $y = C_1e^x + C_2e^x$
16. The auxiliary equation of differential equation $\frac{d^3y}{dx^3} - 7y = 0$ is
- (A) $D^3 - 7y = 0$ (C) $D^3 - 7D = 0$
 (B) $D^3 = 0$ (D) $D^3 - 7 = 0$
17. If $2 + 3i$ and $2 - 3i$ are roots of auxiliary equation of second order homogenous differential equation, then solution of that equation is
- (A) $y = e^{3x}(C_1 \cos 2x + C_2 \sin 2x)$ (C) $y = e^{2x}(C_1 \cos 3x - C_2 \sin 3x)$
 (B) $y = e^{2x}(C_1 \cos 3x + C_2 \sin 3x)$ (D) $y = e^{3x}(C_1 \cos 2x - C_2 \sin 2x)$
18. The Wronskian of two functions $y_1 = \cos x$ and $y_2 = \sin x$ is
- (A) 1 (B) 0 (C) $\cos x$ (D) $\sin x$
19. Which of the following is governing differential equation of population growth and decay model?
- (A) $\frac{dN(t)}{dt} = -kN(t)$ (C) $\frac{d^2N(t)}{dt^2} = kN(t)$
 (B) $\frac{dN(t)}{dt} = kN(t)$ (D) None of above
20. A radioactive substance decomposes at a rate proportional to its mass. This rate is called the decay rate. If $m(t)$ represents the mass of a substance at any time t , then
- (A) $\frac{dm(t)}{dt} = -km(t)$ (C) $\frac{dm^2(t)}{dt^2} = -km(t)$
 (B) $\frac{dm(t)}{dt} = 0$ (D) None of above

SECTION II

Q.2. Attempt any **Four** of the following. [20]

(1) (i) Find 14th derivative of function $y = e^x \sin x$. [03]

(ii) Find 5th derivative of function $y = \cos x$. [02]

(2) Solve the differential equation $\frac{dy}{dx} - 2\frac{y}{x} = x^2$. [05]

(3) Verify Cauchy's mean value theorem for x^3 and $2 - x$, $x \in [0, 9]$. [05]

(4) Solve the differential equation $\frac{d^2y}{dx^2} - 4y = e^{-2x}$. [05]

(5) Solve the differential equation $\frac{dy}{dt} - y = 0$ with initial condition $y(0) = 1$ [05]
using Laplace transform.

(6) State the governing differential equation of Newton's law of cooling and solve it. [05]

SECTION III

Q.3. Attempt any **Four** of the following. [20]

(1) (i) Find 13th derivative of function $y = e^{-8x}$. [02]

(ii) Find 7th derivative of function $y = \log(2x - 3)$. [03]

(2) (i) Find Laplace transform of function $\sin 3t - t^3 + e^{-3t}$. [03]

(ii) Find inverse Laplace transform of function $\frac{1}{s} + \frac{s}{s^2+9} + \frac{10}{s^2+100}$. [02]

(3) Find Taylor's series expansion of $f(x) = e^x$ around $a = 1$. [05]

(4) Solve the differential equation $(y + e^{-x}) dx + (x + 5) dy = 0$. [05]

(5) Solve the differential equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$ [05]

(6) The population of a bacteria is known to increase at a rate proportional to [05]
the number of people present at a time t . If the population has doubled in 6
hours, how long it will take to triple?

Q.4. Attempt any Four of the following.

[20]

(1) (i) Find n^{th} derivative of function $y = x^2 e^x$.

[03]

(ii) Find 11^{th} derivative of function $y = 2^{5x}$.

[02]

(2) Solve the differential equation $x \frac{dy}{dx} + \operatorname{cosec} y = 0$.

[05]

(3) (i) Solve the differential equation $\frac{d^2 y}{dx^2} + y = 0$.

[03]

(ii) Find the Wronskian of two functions $y_1 = 1$ and $y_2 = x$.

[02]

(4) Verify Lagrange's mean value theorem for $f(x) = x^2$, $x \in [1, 3]$.

[05]

(5) State the governing differential equation of radio-active decay and carbon dating and solve it.

[05]

(6) Find inverse Laplace transform of function $\frac{4}{(s-7)^2 + 16}$.

[05]

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